Unit: 6 Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ____________
Date: ________

Advanced (Habanero)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$
(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{4}$
(E) $\frac{1}{4\sqrt{2}}$

Q2. Simplify $\frac{1}{\sqrt{3}+\sqrt{2}}$
(A) $\frac{\sqrt{3}-\sqrt{2}}{1}$
(B) $\frac{\sqrt{3}+\sqrt{2}}{5}$
(C) $\frac{\sqrt{3}-\sqrt{2}}{3-2}$
(D) $\frac{\sqrt{3}-\sqrt{2}}{-1}$
(E) $\frac{\sqrt{3}+\sqrt{2}}{-1}$

Q3. Rationalize the denominator of $\frac{2}{\sqrt{5}-1}$
(A) $\frac{2(\sqrt{5}+1)}{4}$
(B) $\frac{2(\sqrt{5}+1)}{5-1}$
(C) $\frac{2(\sqrt{5}-1)}{5-1}$
(D) $\frac{2(\sqrt{5}+1)}{6}$
(E) $\frac{2(\sqrt{5}-1)}{6}$

Q4. Rationalize the denominator of $\frac{3}{\sqrt{2}+\sqrt{3}}$
(A) $\frac{3(\sqrt{2}-\sqrt{3})}{2-3}$
(B) $\frac{3(\sqrt{2}+\sqrt{3})}{2+3}$
(C) $\frac{3(\sqrt{2}-\sqrt{3})}{2+3}$
(D) $\frac{3(\sqrt{2}+\sqrt{3})}{2-3}$
(E) $\frac{3(\sqrt{3}-\sqrt{2})}{2-3}$

Q5. Simplify $\frac{\sqrt{2}+1}{\sqrt{2}-1}$
(A) $\frac{(\sqrt{2}+1)(\sqrt{2}+1)}{2-1}$
(B) $\frac{(\sqrt{2}+1)(\sqrt{2}-1)}{2-1}$
(C) $\frac{(\sqrt{2}+1)(\sqrt{2}-1)}{1}$
(D) $\frac{(\sqrt{2}-1)(\sqrt{2}+1)}{1}$
(E) $\frac{(\sqrt{2}+1)(\sqrt{2}+1)}{1}$

Q6. Rationalize the denominator of $\frac{1}{\sqrt{6}+\sqrt{2}}$
(A) $\frac{\sqrt{6}-\sqrt{2}}{6-2}$
(B) $\frac{\sqrt{6}+\sqrt{2}}{6+2}$
(C) $\frac{\sqrt{6}-\sqrt{2}}{4}$
(D) $\frac{\sqrt{6}+\sqrt{2}}{4}$
(E) $\frac{\sqrt{6}+\sqrt{2}}{8}$

Q7. Rationalize the denominator of $\frac{1}{\sqrt{3}-1}$
(A) $\frac{\sqrt{3}+1}{3-1}$
(B) $\frac{\sqrt{3}-1}{3-1}$
(C) $\frac{\sqrt{3}+1}{2}$
(D) $\frac{\sqrt{3}-1}{2}$
(E) $\frac{1}{2}$

Q8. Simplify $\frac{2}{\sqrt{5}+1}$
(A) $\frac{2(\sqrt{5}-1)}{4}$
(B) $\frac{2(\sqrt{5}+1)}{4}$
(C) $\frac{2(\sqrt{5}+1)}{5-1}$
(D) $\frac{2(\sqrt{5}+1)}{5+1}$
(E) $\frac{2(\sqrt{5}-1)}{5-1}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{\sqrt{2}+1}{\sqrt{2}-1}$ and simplify.

Q10. Rationalize the denominator of $\frac{1}{\sqrt{6}+\sqrt{3}}$ and simplify.

Chef's Tips

- Tip 1: Make sure to check the denominators for any remaining radicals
- Tip 2: Rationalizing the denominator may involve multiplying the numerator and denominator by a conjugate
- Tip 3: Always simplify your final answer